



AI Clinical Decision Support Lite Hackathon

Day 2

Retrieval Optimization

AI Clinical Decision Support Lite Hackathon

**Inspect
retrieval**



**Tune
Top-K + chunks**



**Measure
Precision@K**

What students will learn and execute during Day 2

Today's learning objectives

Students should leave this session with a measurable retrieval baseline.

Conceptual goal

Understand that the retrieval layer decides what evidence the LLM is allowed to see. Bad retrieval produces unsafe or unsupported answers.

Technical goal

Implement top-k semantic search, inspect retrieved chunks, tune chunk size and overlap, log scores, and optionally add hybrid search or reranking.

Execution goal

Run a small evaluation set, calculate Precision@K, and identify where retrieval fails before moving to generation.

Day 1 handoff check

Before optimization, verify that the knowledge base is ready to search.

Required inputs

- 1–2 official guideline PDFs.
- Clean text extraction.
- Section-aware chunks.
- Embeddings generated.
- Metadata stored with every chunk.

Readiness test

Run 5 clinical questions and display:

- chunk text
- document name
- page number
- section title
- similarity score

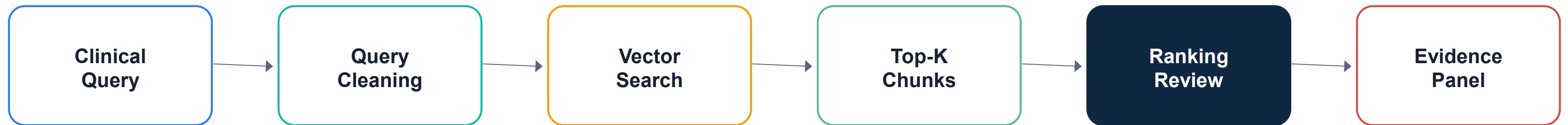
If not ready

Do not start prompt engineering.
Fix parsing, chunks, metadata, or indexing first.

Retrieval optimization only works when the index is trustworthy.

Retrieval pipeline for today

Build a transparent evidence selection layer before answer generation.



Search

Use semantic search to retrieve candidate chunks that are close to the clinical query.

Review

Inspect the top results manually. Scores alone are not enough in clinical use cases.

Display

Show retrieved chunks before generation so the system is explainable to judges.

What makes retrieval “good”?

Good retrieval means the right evidence appears near the top, with enough context to answer.

Relevant

The chunk directly addresses the clinical question, not just a similar medical topic.

Specific

The chunk contains the recommendation, threshold, criteria, or evidence needed for the answer.

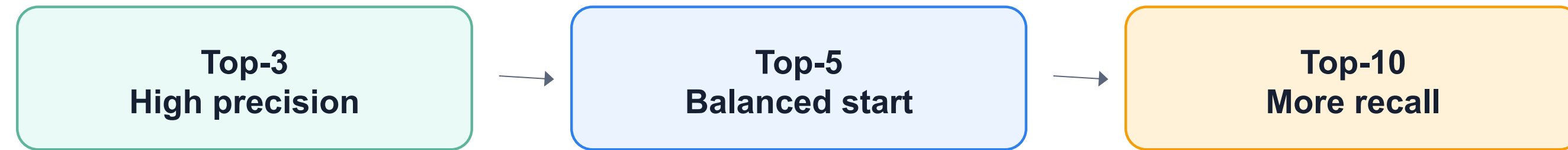
Traceable

The chunk includes page, section, document name, and enough text for a citation.

If the best evidence is not in Top-K, the final answer will likely be weak or unsafe.

Step 1 — Tune Top-K

Top-K controls how many candidate chunks are passed forward for review or generation.



Too small

You may miss the correct evidence, especially when wording differs from the guideline.

Recommended today

Start with Top-5, inspect results, then compare against Top-3 and Top-10.

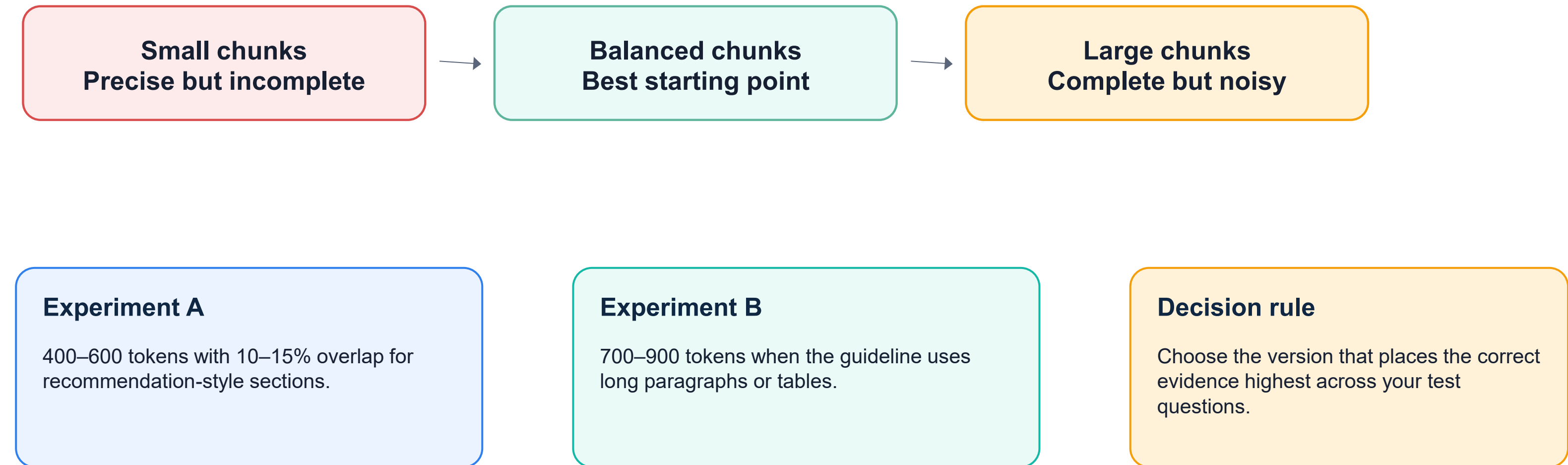
Too large

The LLM may receive noisy or conflicting context unless reranking/filtering is applied.

Step 2 — Tune chunk size and overlap

Retrieval problems often come from chunks, not from the vector database.

Chunk size trade-off



Step 3 — Compare retrieval strategies

Students should understand the difference between semantic, keyword, hybrid, and reranked retrieval.

Semantic search

Finds meaning similarity using embeddings.

Strong for paraphrases.
May miss exact medical terms or abbreviations.

Keyword search

Finds exact terms.

Strong for drug names, acronyms, grades, and thresholds.
Weak for paraphrased questions.

Hybrid search

Combines semantic and keyword signals.

Often better for clinical guidelines with exact terminology.

Reranking

Reorders retrieved candidates using a stronger relevance model.

Good bonus if stable.

Step 4 — Build a mini evaluation set

Retrieval quality must be measured, not guessed.

Create 15–20 questions

Include:

- direct questions
- paraphrased questions
- abbreviation questions
- threshold questions
- out-of-scope questions

Label expected evidence

For each question, record:

- expected document
- page or section
- relevant chunk ID if known
- reason it is relevant

Use for tuning

Run the same questions after each retrieval change.
Keep what improves measurable results.

Do not change many variables at once. Tune one thing, measure, then decide.

Step 5 — Measure Precision@K

Precision@K shows how many retrieved chunks are actually relevant.

Formula

$$\text{Precision@K} = \frac{\text{Relevant retrieved chunks}}{K}$$

Example: if Top-5 contains 3 relevant chunks,
 $\text{Precision@5} = 3 / 5 = 0.60$

Manual validation

A human should mark each retrieved chunk as relevant or not relevant for the question.

Why it matters

This metric directly supports the Retrieval Quality judging category.

Target for demo

Show a small table of questions with Precision@3 and Precision@5.

Step 6 — Design the evidence panel

The interface should make retrieval transparent before the LLM answer appears.

Clinical query: “When should antihypertensive treatment be started?”

Chunk 1 • Score 0.88 • NICE Guideline • Section 4.2 • Page 18

Retrieved evidence excerpt appears here so judges can inspect grounding before generation...

Chunk 2 • Score 0.81 • NICE Guideline • Section 5.2 • Page 19

Retrieved evidence excerpt appears here so judges can inspect grounding before generation...

Chunk 3 • Score 0.74 • NICE Guideline • Section 6.2 • Page 20

Retrieved evidence excerpt appears here so judges can inspect grounding before generation...

UI rule

Common retrieval failure modes

Instructors should help students find these issues early.

Wrong topic

The chunk is medically related but answers a different clinical question.

Fix: improve scope, query cleaning, or reranking.

Missing context

The chunk contains a sentence but not the recommendation criteria around it.

Fix: adjust chunk size or overlap.

Duplicate chunks

Top-K returns almost the same evidence repeatedly.

Fix: deduplicate by section, page, or similarity threshold.

Hands-on lab — Complete this today

Practical checklist for teams after the teaching block.

- Run Top-5 search on 15–20 questions
- Display retrieved chunks with metadata and scores
- Label each retrieved chunk as relevant or not
- Calculate Precision@3 and Precision@5
- Compare at least two chunk settings
- Try Top-3 vs Top-5 vs Top-10
- Optionally test hybrid search or reranking
- Document which setup performs best and why

Stop point: Do not finalize generation prompts until retrieval can surface the correct evidence consistently.

End-of-day review questions

Use these questions to decide whether the team is ready for Day 3: Grounded Generation & Citation.

Retrieval quality

- Does Top-K include the correct evidence?
- Is the best chunk near the top?
- Are scores logged?
- Are weak results visible?

Evaluation quality

- Is there a labeled question set?
- Are Precision@K values calculated?
- Were chunk settings compared?
- Are findings documented?

Demo readiness

- Can the team show query → chunks → citations metadata?
- Can they explain why retrieval is trustworthy?
- Can they show at least one failure case?

Ready for Day 3 when the team can show measured retrieval, not just working search.